LINEAR AND ROBUST CONTROL SYSTEMS

ELEC 9731

1 - Matrix Methods Review

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Topics

1 Matrix and Vector - Basic Properties

2 Linear Dependence and Rank

3 Eigenvalues and Eigenvectors:
Eigenvalue Decomposition (EVD) and
Singular Value Decomposition (SVD)

4 Functions of Matrices

5 Partitioned Matrix Inversion

6 Generalised Inverses

7 Examples

References

G. Strang. Applied Linear Algebra.

T. Kailath. Linear Systems. Appendices.

(A) Matrix and Vector - Basic Properties

Matrix (Rectangular)

$A = A_{m \times n} = [a_{ij}]$; m rows, n columns

Column vector: $x = x_{n \times 1} = [x_i]$

Adjoint: $A^T = A^T_{n \times m} = [a_{ji}]$

Conformable product

$C = AB \leftrightarrow c_{ij} = \sum_k a_{ik} b_{kj}$

Inner product

$a^T b = \sum_1^n a_i b_i$

Outer product (dyadic product)

$c_{m \times 1} d^T_{p \times 1} = [c_i d_j]_{m \times n}$

Positive Definite Matrix

$x^T A x > 0, \underline{x} \neq \underline{0}$

Positive Semi-Definite Matrix

$x^T A x \geq 0$

Symmetric Matrix

$A = A^T$

Hermetian transpose = complex conjugate transpose

$A^H = (A^*)^T$

(A) Matrix and Vector - Basic Properties - Continued

<u>Norms</u>

$$\begin{aligned} \| x \| &= \| x \|_2 = \sqrt{x^T x} = \sqrt{\sum_1^n x_i^2} \\ \| x \|_p &= (\sum_1^n |x_i|^p)^{\frac{1}{p}} \end{aligned}$$

Angle between vectors

$\cos \theta_{x,y} = \frac{x^T y}{\|x\| \|y\|}$

Orthogonal vectors

$\theta_{x,y} = \frac{\Pi}{2}$ or $x^T y = 0$

(B) Linear Dependence and Rank

[B1] Linear dependence and vector spaces

A set of m-vectors $c_1, \cdots, c_n$ is said to be linearly independent if there is no linear combination $\sum_1^n c_i v_i$ that is zero.
The vector space spanned by $c_1, \cdots, c_n$ is the set of vectors obtainable from $c_1, \cdots, c_n$ by addition and scalar multiplication.

Collect $c_1, \cdots, c_n$ into a matrix $A_{n\times m}$. Then note that we can express A in terms of its column vectors or row vectors

$$A_{m\times n} = [c_1, \cdots, c_n] = \begin{bmatrix} r_1^T \\ \vdots \\ \vdots \\ r_m^T \end{bmatrix}$$

(B) Linear Dependence and Rank - Continued

Forming linear combinations of columns of A is equivalent to multiplying A into a vector since

$$Av = [c_1, \cdots, c_n] \begin{bmatrix} v_1 \\ \vdots \\ \vdots \\ v_n \end{bmatrix} = \sum_1^n c_i v_i$$

Similarly vector pre multiplication generates linear conbinations of row vectors

$$w^T A = [w_1, \cdots, w_m] \begin{bmatrix} r_1^T \\ \vdots \\ \vdots \\ r_m^T \end{bmatrix} = \sum_1^m w_j r_j^T$$

(B) Linear Dependence and Rank - Continued II

[B2] The four fundamental sub spaces

(i) Column space

The column (vector) space of A is the vector space spanned by the columns of A denoted $cs(A)$. It is also called the range of A and denoted $R(A)$.

(ii) Row space

The row space is the space spanned by the rows. Since $(w^T A)^T = A^T w$ it is also the column space of A^T and so denoted $cs(A^T)$ or $R(A^T)$.

(iii) Right Null space

or kernel of A is the space spanned by the null vectors v_r such that $Av_r = 0$. It is denoted $N(A)$ or $ker(A)$.

(iv) Left Null space

or left kernel is the space spanned by the null vectors w_k such that $w_k^T A = 0$. Since $(w_k^T A)^T = A^T w_k$ this is also the right null space of A^T and denoted $N(A^T)$.

(B) Linear Dependence and Rank - Continued III

[B3] Rank & Nullity

$$
\begin{aligned}
r &= \text{(column) rank of} A \\
&= \text{dim. column space of} A \\
&= dim.cs(A) = dim.R(A) \\
\nu &= \text{(column) nullity of} A \\
&= \text{dim. right null space of} A \\
&= dim.N(A)
\end{aligned}
$$

Clearly $r = \#$ linearly independent columns

$\nu = \#$ dependent columns

So $r + \nu = n$

Similarly

r_T = row rank of A = dim $R(A^T)$

ν_T= row nullity of A = dim $N(A^T)$

and $r_T + \nu_T = m$

(B) Linear Dependence and Rank - Continued IV

[B4] Fundamental Theorem of Matrix Algebra

We show every n-vector is in $N(A)$ or $R(A^T)$

If this holds then $\nu + r_T = n$ but
$\nu + r = n \Rightarrow r = r_T$ i.e. column rank=row rank and this is the fundamental result.

The result is obvious once we write $Av = 0$ as

$$\begin{pmatrix} r_1^T \\ \vdots \\ r_m^T \end{pmatrix} v = 0 \leftrightarrow r_i^T v = 0$$

So every null vector is $\perp$ to every row vector (and linear combinations thereof) and any vector $\perp$ to every row vector is automatically a null vector.

Thus the rank of a matrix is its common row and column rank.

If A is a square matrix we say it has <u>full rank</u> of rank = order. It is then called non-singular. Otherwise it is singular.

- Continued V

[B5] Determinants.

A has a determinant iff it is square. If a linear combination of columns is added to another column the determinant is unchanged. So if A is singular we can make one of its columns all $0's$. Then it has zero determinant. So det $A = |A| = 0$ iff A is singular.

[B6] Inverse

A has a left inverse A_L if $A_L A = I$. A has a right inverse A_R if $AA_R = I$

If A is square it has an inverse iff it has full rank. [Suppose $A^{-1}A = I$ but for some $c \neq 0$, $Ac = 0 \Rightarrow A^{-1}0 = c = 0$ a contradiction].

N.B. A^{-1} =adjoint(A)/$detA$

$(ABC)^{-1} = C^{-1}B^{-1}A^{-1}$

(C) Eigenvalues and Eigenvectors

[C1] If $A_{n\times n}$ is a square matrix, a vector p satisfying $Ap = \lambda p$ for some scalar λ is called a right eigenvector of A and λ is called an eigenvalue (eval).

[C2] Similarly a left eigenvector satisfies

$q^T A = \lambda' q^T \leftrightarrow A^T q = \lambda' q$

and λ' is another eigenvalue.

[C3] Left and right eigenvectors are orthogonal for distinct eigenvalues.

Proof: Let q be a left eigenvector with eval λ.

Let p be a right evec with eval λ'

$$\begin{aligned}
\Rightarrow q^T Ap &= \lambda q^T p \\
&= \lambda' q^T p \\
\Rightarrow q^T p &= 0 \text{ if} \lambda \neq \lambda'
\end{aligned}$$

(C) Eigenvalues and Eigenvectors - Continued

[C4] Finding eigenvalues

If A is square we must solve $(A - \lambda I)p = 0$ to find λ. This means $A - \lambda I$ is singular and this implies $|A - \lambda I| = 0$

Now this equation is an nth degree polynomial in λ and so has n roots which are the eigenvalues. Thus there may be repeated eigenvalues. This polynomial is the characteristic polynomial of A.

[C5] (i) Neither p nor q need be unit vectors when $p^T q = 0$

(ii) If A is symmetric the evals are real since

$$\begin{aligned} Ap &= \lambda p \Rightarrow Ap^* = \lambda^* p^* \\ \Rightarrow p^H Ap &= \lambda p^H p \\ \text{also } p^H Ap &= \lambda^* p^H p \\ \Rightarrow \lambda &= \lambda^* \end{aligned}$$

(C) Eigenvalues and Eigenvectors - Continued II

[C6] Eigenvalue Decomposition (EVD)

(also called spectral decomposition).

Suppose the evals are distinct : $\lambda_1, \cdots, \lambda_n$

Let $P = (p_1, \cdots, p_n), Q = (q_1, \cdots, q_n)$ be the corresponding right and left evecs. These are mutually orthogonal so

$$\begin{aligned} Q^T P &= [q_i^T p_j] = [\delta_{ij}] = I \\ \Rightarrow Q^T &= P^{-1} \end{aligned}$$

(The p_i are linearly independent so p has full rank and so has an inverse). Continuing

$$\begin{aligned} \Rightarrow PQ^T &= I \\ \Rightarrow (p_1, \cdots, p_n) \begin{pmatrix} q_1^T \\ \vdots \\ q_n^T \end{pmatrix} &= I \\ \Rightarrow \sum_1^n p_i q_i^T &= I \end{aligned}$$

(C) Eigenvalues and Eigenvectors - Continued III

From this we can get the EVD.

$$
\begin{aligned}
Ap_i &= \lambda_i p_i, i = 1, \cdots, n \\
\Rightarrow Ap_i q_i^T &= \lambda_i p_i q_i^T, i = 1, \cdots, n \\
\Rightarrow A \sum_1^n p_i q_i^T &= \sum_1^n \lambda_i p_i q_i^T \\
\Rightarrow A &= P \Lambda Q^T = \sum_1^n \lambda_i p_i q_i^T
\end{aligned}
$$

(C) Eigenvalues and Eigenvectors - Continued IV

[C7] Cayley Hamilton Theorem

From the EVD

$A^2 = P\Lambda Q^T P\Lambda Q^T = P\Lambda^2 Q^T$

induction $\Rightarrow A^m = P\Lambda^m Q^T$

Now suppose $|A - \lambda I| = \sum_1^n a_u \lambda^u$

<u>Cayley Hamilton Theorem</u>: Then $\sum_1^n a_u A^u = 0$

(C) Eigenvalues and Eigenvectors - Continued V

Proof:

$$\begin{aligned}
\sum_{1}^{n} a_u A^u &= \sum_{1}^{n} a_u P \Lambda^u Q^T \\
&= P(\sum_{1}^{n} a_u \Lambda^u) Q^T \\
&= P diag(\sum_{1}^{n} a_u \lambda_1^u, \cdots, \sum_{1}^{n} a_u \lambda_n^u) Q^T \\
&= P diag(0, \cdots, 0) Q^T \\
&= 0
\end{aligned}$$

NB

(i) If A is symmetric then left evecs=right evecs so $Q = P$.

(ii) Again note very carefully that in the EVD even if p_i have unit length q_i will not (and vice versa). We have only $p_i^T q_i = 1$.

(C) Eigenvalues and Eigenvectors - SVD

[C8] Singular Value Decomposition (SVD)

$A_{m\times n}$ is rectangular $m > n$

So rank $(A) \leq n$

Then $AA^T_{m\times m}$, $A^T A_{n\times n}$ are square and AA^T has rank $\leq n$, so it has at least $m-n$ null vectors $\Rightarrow m-n$ zero evals. Next

$$\left|\lambda I - AA^T\right| = |\lambda I - A^T A|$$

so the remaining n evals of AA^T are the same as those of $A^T A$.

Suppose these n evals are distinct. Since $A^T A$ is positive semidefinite these are non negative so we denote them $\rho_1^2, \cdots, \rho_n^2$.

(C) Eigenvalues and Eigenvectors - SVD Continued II

Let $u_1, \cdots, u_n$ be the corresponding evecs. They are orthogonal (we make them unit length),

$$\begin{aligned} A^T A u_i &= \rho_i^2 u_i, i = 1, \cdots, n & (0.1) \\ \text{let} \quad v_i &= A u_i / \rho_i, \ (\text{so} v_i^T v_i = 1) & (0.2) \\ \Rightarrow A^T v_i &= A^T A u_i / \rho_i = \rho_i u_i & (0.3) \end{aligned}$$

Then consider that further

$$A A^T v_i = A u_i \rho_i = \rho_i^2 v_i \qquad (0.4)$$

So v_i are evecs of AA^T with evals ρ_i^2 and they are mutually orthogonal.

We can now establish the
<u>singular value decomposition</u> of A

$$\text{SVD:} \quad A = \sum_1^n \rho_i v_i u_i^T = V \Lambda U^T \qquad (0.5)$$

(not to be confused with the EVD : they agree when A is symmetric.)

(C) Eigenvalues and Eigenvectors - SVD Continued III

Consider the $n \times n$ matrix

$$\begin{aligned}
& (A - \sum_1^n \rho_i v_i u_i^T)^T (A - \sum_1^n \rho_i v_i u_i^T) \\
= \quad & (A^T - \sum_1^n \rho_i u_i v_i^T)(A - \sum_1^n \rho_i v_i u_i^T) \\
= \quad & A^T A - \sum_1^n \rho_i^2 u_i u_i^T \\
- \quad & \sum_1^n \rho_i^2 u_i u_i^T + \sum_1^n \rho_i^2 u_i u_i^T \\
= \quad & A^T A - \sum_1^n \rho_i^2 u_i u_i^T \\
= \quad & 0 \text{ by EVD}
\end{aligned}$$

This establishes the result.

(C) Eigenvalues and Eigenvectors - SVD Continued IV

To summarise

(i) (0.2) $\Rightarrow Au_i = \rho_i v_i \leftrightarrow A^T A u_i = \rho_i^2 u_i$

(ii) (0.4) $\Rightarrow A^T v_i = \rho_i u_i \leftrightarrow A A^T v_i = \rho_i^2 v_i$

compare to EVD

$Ap_i = \lambda_i p_i, A^T q_i = \lambda_i q_i, A = \sum_1^n \lambda_i p_i q_i^T$

when A is symmetric $v_i = u_i$ and $\rho_i = \lambda_i$

(C) Eigenvalues and Eigenvectors - SVD Continued V

(iii) SVD tells about near singularity. Suppose one singular value ρ_0 is small. Then consider the relation:

$$\begin{aligned} Au_0 &= \rho_0 v_0 \\ \leftrightarrow (c_1, \cdots, c_n) \begin{pmatrix} u_{01} \\ \vdots \\ u_{on} \end{pmatrix} &= \rho_0 v_0 \\ \Rightarrow \sum_1^n c_i u_{oi} &= \rho_0 v_0 \end{aligned}$$

Now v_0 is a unit vector so this equation gives a linear combination of columns of A that is nearly 0.

Similarly the equation $A^T v_0 = \rho_0 u_0$ gives a linear combination of rows of A that is nearly zero.

D. Matrix Functions

We already defined $f(A) = Pf(\Lambda)Q^T$

[D1] Matrix exponential can then be defined in several ways. With distinct eigenvalues

$$
\begin{aligned}
e^{At} &= Pe^{\Lambda t}Q^T = P\ \mathrm{diag}(e^{\lambda_i t})Q^T \\
&= P\ \mathrm{diag}(\sum_0^\infty \frac{(\lambda_i t)^r}{r!})Q^T \\
&= \sum_0^\infty \frac{t^r}{r!} P\mathrm{diag}(\lambda_i^r)Q^T \\
&= \sum_0^\infty \frac{t^r}{r!} A^r
\end{aligned}
$$

e.g. $A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \Rightarrow A^2 = 0 \Rightarrow A^m = 0,\ m \geq 2$

$\Rightarrow e^{At} = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}$

If A has repeated ergenvalues $A^m = PJ^mQ^T$ etc.. J is the Jordan form.

[D2] Matrix Calculus

(a) Matrices

If $A(t) = [a_{ij}(t)]$ can then easily show

$\frac{d}{dt}A(t)B(t) = A(t)\frac{dB}{dt} + B(t)\frac{dA}{dt}$

$\frac{d}{dt}A^{-1} = -A^{-1}\frac{dA}{dt}A^{-1}$

Similarly

$$\begin{aligned}
\mathcal{L}(A(t)) &= \overline{A}(s) = \int_0^\infty A(t)e^{-st}dt \\
&= [\int_0^\infty a_{ij}(t)e^{-st}dt] \\
&= [\overline{a}_{ij}(s)]
\end{aligned}$$

e.g. $\mathcal{L}(e^{At}) = (sI - A)^{-1}$

[D2] Matrix Calculus - Continued

(b) Vectors

If $\phi = f(x) = f(x_1, \cdots, x_n)$

Then $\frac{d\phi}{dx} = \left(\frac{\partial f}{\partial x_1}, \cdots, \frac{\partial f}{\partial x_n}\right)^T$

If $\underline{\phi} = \underline{f}(x)$, then

$$\frac{d\underline{\phi}}{d\underline{x}}^T = \begin{bmatrix} \frac{\partial \phi_1}{\partial x_1} & \frac{\partial \phi_2}{\partial x_1} & \cdots & \frac{\partial \phi_m}{\partial x_1} \\ \vdots & & & \\ \vdots & & & \\ \vdots & & & \\ \frac{\partial \phi_1}{\partial x_n} & \cdots & \cdots & \frac{\partial \phi_n}{\partial x_n} \end{bmatrix}$$

Then can show

$$\begin{aligned} \frac{d}{d\underline{x}} c^T \underline{x} &= \underline{c} \\ \frac{d}{d\underline{x}} (A\underline{x})^T &= A^T \\ \frac{d}{d\underline{x}} x^T A x &= Ax + A^T x = 2Ax (A symmetric) \end{aligned}$$

[D2] Matrix Calculus - Continued II

[D3] Matrix Integration

$\int_0^T A(s)ds = \int_0^T a_{ij}(s)ds]$

[D4] Signal Norms

$$\begin{aligned} \| \, s \, \|_2 &= (\int_{-\infty}^{\infty} s^2(t)dt)^{\frac{1}{2}} \\ \| \, s \, \|_p &= (\int_{-\infty}^{\infty} |s(t)|^p \, dt)^{1/p} \uparrow \| \, s \, \|_\infty = \sup_t |s(t)| \end{aligned}$$

[E] MATRIX INVERSION

[E1] Partitioned Matrix Inversion

If $\underline{A}$,$\underline{B}$ are square matrices and the indicated inverses exist, then

$$\begin{pmatrix} \underline{A} & \underline{D} \\ \underline{B} & \underline{C} \end{pmatrix}^{-1} =$$

$$\begin{bmatrix} \underline{A}^{-1} + \underline{A}^{-1}\underline{D}\underline{\Delta}_c^{-1}\underline{B}\underline{A}^{-1} & -\underline{A}^{-1}\underline{D}\underline{\Delta}_c^{-1} \\ -\underline{\Delta}_c^{-1}\underline{B}\underline{A}^{-1} & \underline{\Delta}_c^{-1} \end{bmatrix}$$

where $\underline{\Delta}_c = \underline{C} - \underline{B}\underline{A}^{-1}\underline{D}$

The result is easily checked by direct multiplication. In a similar way show

$$\begin{pmatrix} \underline{A} & \underline{D} \\ \underline{B} & \underline{C} \end{pmatrix}^{-1} =$$

$$\begin{bmatrix} \underline{\Delta}_A^{-1} & -\underline{\Delta}_A^{-1}\underline{D}\underline{C}^{-1} \\ -\underline{C}^{-1}\underline{B}\underline{\Delta}_A^{-1} & \underline{C}^{-1} + \underline{C}^{-1}\underline{B}\underline{\Delta}_A^{-1}\underline{C}^{-1} \end{bmatrix}$$

where $\underline{\Delta}_A = \underline{A} - \underline{D}\underline{C}^{-1}\underline{B}$

[E] MATRIX INVERSION - Continued

[E2] Matrix Inversion Lemma

Equating the two inverses in section E1 gives

$(\underline{A} - \underline{DC}^{-1}\underline{B})^{-1} = \underline{A}^{-1} + \underline{A}^{-1}\underline{D\Delta}_c^{-1}\underline{BA}^{-1}$

or in a more usual notation (change $-\underline{C}^{-1}$ with $\underline{C}$ and interchange $\underline{B}$ and $\underline{D}$)

$(\underline{A} + \underline{BCD})^{-1} = \underline{A}^{-1} - \underline{A}^{-1}\underline{B\Delta}^{-1}\underline{DA}^{-1}$

$\underline{\Delta} = \underline{C}^{-1} + \underline{DA}^{-1}\underline{B}$

[E] MATRIX INVERSION - Continued II

[E3] Determinant Identities

Expand in rows to see that

$$\det \begin{bmatrix} \underline{A} & \underline{D} \\ \underline{0} & \underline{C} \end{bmatrix} = \det \underline{A} \det \underline{C} = \det \begin{bmatrix} \underline{A} & \underline{0} \\ \underline{B} & \underline{C} \end{bmatrix}$$

Now use the identity

$$\begin{pmatrix} \underline{A} & \underline{D} \\ \underline{B} & \underline{C} \end{pmatrix} = \begin{pmatrix} \underline{A} & \underline{0} \\ \underline{B} & \underline{I} \end{pmatrix} \begin{pmatrix} \underline{I} & \underline{A}^{-1}\underline{D} \\ \underline{0} & \underline{\Delta}_c \end{pmatrix}$$

to find that (with $\underline{\Delta}_c$ as given in section E1)

$$det \begin{pmatrix} \underline{A} & \underline{D} \\ \underline{B} & \underline{C} \end{pmatrix} = \det \underline{A} \det \underline{\Delta}_c$$

Similarly show that this also is

$$\det \begin{pmatrix} \underline{A} & \underline{D} \\ \underline{B} & \underline{C} \end{pmatrix} = \det \underline{C} \det \underline{\Delta}_A$$

Now set $\underline{A} = \underline{I}_m, \underline{C} = \underline{I}_n$ to see from the last two identities that

$$\det \left(\underline{I}_n - \underline{BD} \right) = \det(\underline{I}_m - \underline{DB})$$

In particular, if $\underline{b}, \underline{d}$ are vectors,

$$\begin{aligned} det(\underline{A} + \underline{bd}^T) &= \det[(\underline{A})(\underline{I} + \underline{A}^{-1} bd^T)] \\ &= \det(\underline{A}) \det(\underline{I} + \underline{A}^{-1} \underline{bd}^T) \\ &= (1 + \underline{d}^T \underline{A}^{-1} \underline{b}) \det \underline{A} \end{aligned}$$

F. Generalised Inverses

[Fi)] We say $G^{-}_{m\times p}$ is a generalised inverse (g-inverse) of $G_{p\times m}$ if $GG^{-}G = G$

Example $G = ab^T$, then $G^{-} = \frac{ba^T}{a^T ab^T b}$

Proof:

$GG^{-}G = ab^T \frac{ba^T}{a^T ab^T b} ab^T = ab^T = G$

[F(ii)] solving $Gx = b$ when G is singular.

Suppose G has rank r and let $\phi_1, \cdots, \phi_k$ be linearly independent null vectors i.e.

$G\phi_t = 0, t = 1, \cdots, k$

If G^{-} is any g-inverse of G, then a general solution to $Gx = b$ is (assuming there is one)

$x = G^{-}b + \sum_1^k \alpha_r \phi_r$

where α_r are free parameters.

F. Generalised Inverses - Continued

Proof. Let x_0 be a solution so that $Gx_0 = b$, then

$Gx = GG^{-}b + 0 = GG^{-}Gx_0 = Gx_0 = b$

Let $x_1$be another solution so that $Gx_1 = b$, then

$$
\begin{aligned}
G(x_1 - x_0) &= b - b = 0 \\
\Rightarrow x_1 - x_0 &= \sum_1^k \theta_r \phi_r \text{ for some } \theta_r^{'} s \\
\Rightarrow x_1 &= x_0 + \sum_1^k \theta_r \phi_r \\
&= G^{-}b + \sum_1^k \alpha_r \phi_r
\end{aligned}
$$

recovering the general form.

F. Generalised Inverses - Continued II

[F(iii)] Moore - Penrose g-inverse

Let G have SVD

$$\underset{p\times m}{G} = \underset{p\times n}{U} \quad \underset{n\times n}{\Lambda} \quad \underset{n\times m}{V^T}$$

$n =$rank$G \leq \min(p, m)$. Note that

$U^T U = I_n = V^T V$

Then a g-inverse is

$$\underset{mx\times p}{G^+} = \underset{m\times n}{V} \quad \underset{n\times n}{\Lambda^{-1}} \quad \underset{n\times p}{U^T}$$

It is called the Moore-Penrose g-inverse

<u>Proof</u>

$$\begin{aligned} GG^+G &= U\Lambda V^T(V\Lambda^{-1}U^T)U\Lambda V^T \\ &= U\Lambda\Lambda^{-1}\Lambda V^T \\ &= U\Lambda V^T = G \end{aligned}$$

F. Generalised Inverses - Continued III

[F(iv)] Moore-Penrose Properties

$$\begin{aligned}
G^+G &= V\Lambda^{-1}U^TU\Lambda V^T \\
&= V\Lambda^{-1}\Lambda V^T \\
&= VV^T_{m\times m}(\neq I_m) \\
GG^+ &= U\Lambda V^TV\Lambda^{-1}U^T \\
&= U\Lambda\Lambda^{-1}U^T \\
&= UU^T_{p\times p}(\neq I_p)
\end{aligned}$$

[F(v)] Projection g-inverse

Suppose $p > m$; if G has rank m then a g-inverse is

$$G^- = (G^TG)^{-1}G^T$$

Proof: $GG^-G = G(G^TG)^{-1}G^TG = G$

F. Generalised Inverses - Continued IV

[F(vi)] Projection Matrices

Suppose $X_{n\times r}$ has rank $r(< n)$. Find $X_\perp (n-r)\times n$ of rank $n-r$, orthogonal to X

i.e. $X_\perp^T X = 0$

i.e. all columns of $X_\perp$ are orthogonal to all columns of X and $W = [X, X_\perp]$ has full rank.

<u>Projection Lemma</u>:
$I_{n\times n} = X(X^TX)^{-1}X^T + X_\perp(X_\perp^T X_\perp)^{-1}X_\perp^T$

Proof: Let

$$\begin{aligned}
M &= \begin{bmatrix} (X^TX)^{-1} & X^T \\ (X_\perp^T X_\perp)^{-1} & X_\perp^T \end{bmatrix} \\
\Rightarrow MW &= \begin{pmatrix} (X^TX)^{-1}X^T \\ (X_\perp^T X_\perp)^{-1}X_\perp^T \end{pmatrix} (X, X_\perp) \\
&= \begin{pmatrix} I_{r\times r} & 0 \\ 0 & I_{(n-r)\times(n-r)} \end{pmatrix}
\end{aligned}$$

F. Generalised Inverses - Continued V

$$
\begin{aligned}
&= I_{n\times n} \\
\Rightarrow W &= M^{-1} \\
\Rightarrow WM &= I \\
&= (X, X_\perp)\begin{pmatrix} (X^TX)^{-1}X^T \\ (X_\perp^T X_\perp)^{-1}X_\perp^T \end{pmatrix} \\
&= X(X^TX)^{-1}X^T + X_\perp(X_\perp^T X_\perp)^{-}X_\perp^T
\end{aligned}
$$

as required

N.B. $P_X = X(X^TX)^{-1}X^T$ is a projection matrix

e.g. if

$$
\begin{aligned}
\xi &= Xa + X_\perp b \\
\Rightarrow P_X\xi &= X(X^TX)^{-1}X^Ta + 0 \\
&= X\overline{a} \\
&\in \text{ column space of } X
\end{aligned}
$$

F. Generalised Inverses - Continued VI

[F(vii)] Miscellaneous

(a) Column interchange is represented by post multiplication by a permutation matrix

e.g. $\begin{pmatrix} a_1 & b_1 \\ a_2 & b_2 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} b_1 & a_1 \\ b_2 & a_2 \end{pmatrix}$

(ii) If the cols of Y are linearly dependent on the cols of X then $Y = XA$ for some matrix A

Proof. If v is linearly dependent on $\xi_1, \cdots, \xi_r$, the columns of X then $v = \sum_1^r \alpha_r \xi_r$, for some coefficients α_r

$$\Rightarrow v = (\xi_1, \cdots, \xi_r) \begin{pmatrix} \alpha_1 \\ : \\ \alpha_r \end{pmatrix} = X\alpha$$

So there are vectors $a_1, \cdots, a_m$ with

$$\begin{aligned} Y &= (y_1, \cdots, y_m) \\ &= (Xa_1, \cdots, Xa_m) \\ &= X(a_1, \cdots, .a_m) = XA \end{aligned}$$

G. Examples

[G(i)] Linear Combination of Columns

$$A = \begin{pmatrix} 3 & 1 & 2 \\ 2 & 2 & 1 \\ 1 & 3 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ -2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$= 1 \times \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix} + 1 \times \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} - 2 \times \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix}$$

[G(ii)] 2x2 Matrix Inverse

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{\Delta} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

$$\Delta = det = ad - bc$$

[G(iii)] Evals and Evecs

$$A = \begin{pmatrix} 2 & 2 \\ 3 & 1 \end{pmatrix}$$

G. EXAMPLES - Continued

For evals solve

$$
\begin{aligned}
|\lambda I - A| &= \begin{vmatrix} \lambda - 2 & -2 \\ -3 & \lambda - 1 \end{vmatrix} \\
&= (\lambda - 2)(\lambda - 1) - 6 \\
&= \lambda^2 - 3\lambda + 2 - 6 \\
&= \lambda^2 - 3\lambda - 4 \\
&= (\lambda - 4)(\lambda + 1)
\end{aligned}
$$

$\Rightarrow$ Roots are $\lambda_1, \lambda_2 = 4, -1$

For right evecs solve: $Ap_i = \lambda_i p_i i = 1, 2$

For λ_1

$$
\begin{pmatrix} 2 & 2 \\ 3 & 1 \end{pmatrix} \begin{pmatrix} p_1^{(1)} \\ p_2^{(1)} \end{pmatrix} = 4 \begin{pmatrix} p_1^{(1)} \\ p_2^{(1)} \end{pmatrix}
$$

$$\Rightarrow 2p_1^{(1)} + 2p_2^{(1)} = 4p_1^{(1)}$$

$$3p_1^{(1)} + p_2^{(1)} = 4p_2^{(1)}$$

$$\Rightarrow 2p_2^{(1)} = 2p_1^{(1)} \Rightarrow p_2^{(1)} = p_1^{(1)}$$

Second equation agrees.

G. EXAMPLES - Continued II

We may take

$$\begin{pmatrix} p_1^{(1)} \\ p_2^{(1)} \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

For λ_2

$$\begin{pmatrix} 2 & 2 \\ 3 & 1 \end{pmatrix} \begin{pmatrix} p_1^{(2)} \\ p_2^{(2)} \end{pmatrix} = -1 \times \begin{pmatrix} p_1^{(2)} \\ p_2^{(2)} \end{pmatrix}$$

$$\Rightarrow 2p_1^{(2)} + 2p_2^{(2)} = -p_1^{(2)}$$

$$\Rightarrow 3p_1^{(2)} + p_2^{(2)} = -p_2^{(2)}$$

$$\Rightarrow 2p_2^{(2)} = -3p_1^{(2)}$$

$$\Rightarrow p_2^{(2)} = -\frac{3}{2}p_1^{(2)}$$

and second equation agrees.

We may take $\begin{pmatrix} p_1^{(2)} \\ p_2^{(2)} \end{pmatrix} = \begin{pmatrix} 2 \\ -3 \end{pmatrix}$

G. EXAMPLES - Continued III

So

$$\begin{aligned} P &= \begin{pmatrix} 1 & 2 \\ 1 & -3 \end{pmatrix} \\ \Rightarrow Q^T &= P^{-1} \\ &= \frac{1}{-5} \begin{pmatrix} -3 & -2 \\ -1 & 1 \end{pmatrix} \\ &= \begin{pmatrix} \frac{3}{5} & \frac{2}{5} \\ \frac{1}{5} & -\frac{1}{5} \end{pmatrix} \\ &= \begin{pmatrix} q_1^T \\ q_2^T \end{pmatrix} \end{aligned}$$

which gives left evecs

G. EXAMPLES IV

Use the same matrix as above $A = \begin{pmatrix} 2 & 2 \\ 3 & 1 \end{pmatrix}$

and find its SVD. We have

$$\begin{aligned} A^T A &= \begin{pmatrix} 2 & 3 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} 2 & 2 \\ 3 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 8 & 8 \\ 8 & 10 \end{pmatrix} \end{aligned}$$

Repeating the type of calaulations above gives roots

$\rho_1 = 9 + 2\sqrt{5}; \rho_2 = 9 - \sqrt{5}$ etc...

Symbolic Computation in Matlab

e.g. given $A = \begin{pmatrix} a & b \\ 0 & c \end{pmatrix}, x = \begin{pmatrix} 1 \\ -d \end{pmatrix}$.

Find Ax

Matlab code is

```
>> syms a
>> syms b
>> syms c
>> syms d
```

```
A*b
```

```
ans =
```

```
[ a-b*d, -c*d]
```